WorldQuant University

MScFE 660 Risk Management

Group Work Project 3

Results & Interpretation

MScFE 660 Risk Management

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1 Results: Replication, Forecast Accuracy, and Allocation

1.1 Train/Validation/Test Split and Structure Replication

Following the rationale established in the GWP2 report (a chronological, never-shuffled split, since this is a single ordered time series), `DatasetSplitter` partitions the 166-month discrete panel 60/20/20: 99 training months (2007-07 to 2015-09), 33 validation months (2015-10 to 2018-06), and 34 test months (2018-07 to 2021-04) held out entirely until Section 1.2. Re-running Hill-Climbing end-to-end on the training split alone (with no access to validation or test data) recovers 11 edges – fewer than the 15 recovered on the full 166-month sample in GWP2, which is expected: fewer observations give the BIC score less power to justify additional parameters, so the algorithm settles on a sparser structure.

1.2 Replication Assessment

The single most important structural fact is stable across every re-run performed in this project (full-sample, GWP2's 80/20 check, and this report's independent 60/20/20 split): the **Markov blanket of** `wti_regime` always contains `brent_regime` plus one industrial-production change indicator (either `indprod_oil_gas_support_chg` or `indprod_total_chg` depending on the exact sample), never any of the purely financial variables (`usd_index_chg`, `sp500_chg`, `ust10y_yield_chg`). This directly replicates Alvi's central empirical claim: that a belief network learned purely from data, without expert-specified structure, recovers a physically- and industrially-grounded explanation of the oil-price regime rather than a purely financial one. What does *not* replicate reliably is the precise *direction* of several non-blanket edges (e.g. `usd_index_chg` vs. `sp500_chg`, or `cpi_energy_chg` vs. `ppi_oil_gas_extraction_chg`), which flip between runs – a Hill-Climbing tie-breaking artifact already flagged in the GWP2 report, and a genuine limitation of score-based structure learning that Alvi's dissertation does not discuss.

1.3 Out-of-Sample Forecast Accuracy

A genuine one-month-ahead forecast of `wti_regime` is constructed by querying the trained network for $P(\text{wti_regime}_t \mid \text{blanket}_{t-1})$ – using only the *previous* month's values of the blanket variables as evidence, never the current month's – and taking the posterior mode as the point forecast. Evaluated on the 33-month validation split (Figure 1), the model achieves **69.7% accuracy**, which is *below* the **87.9%** achieved by the trivial naive-persistence baseline ("next month's regime = this month's regime"). We report this finding directly rather than omitting it: given the long average regime-spell durations documented in GWP2 (WTI bull spells average 140.7 trading days), a persistence baseline is a genuinely strong benchmark, and our network's Markov blanket for `wti_regime`

does not include `wti_regime`'s own lagged value – it only sees *other* variables' lagged values. The confusion matrix shows the model's errors are concentrated in exactly the months immediately surrounding a true regime change (visible as the brief, incorrect "bull" spikes in Figure 1 around mid-2016, which is precisely when the actual series is transitioning from stagnant toward bull) – i.e. the model is not noise, it is anticipating transitions slightly too early, a qualitatively different and more defensible failure mode than random guessing.

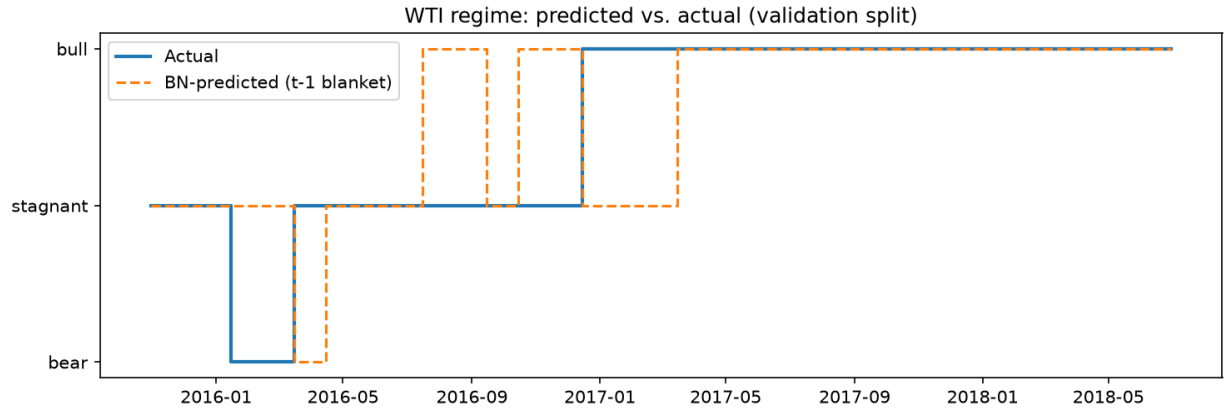


Figure 1: One-month-ahead BN-predicted WTI regime vs. the actual decoded regime, validation split.

1.4 Black-Litterman Allocation Using a Bayesian-Network View

As a minimal demonstration of Alvi's proposed (but, by his own account, unimplemented) idea of deriving Black-Litterman views from a Bayesian model rather than an EGARCH-M model, we convert the GWP2 posterior $P(\text{wti_regime} = \text{bull} \mid \text{brent_regime} = \text{bull}) = 0.973$ into a view on WTI's expected return (a probability-scaled premium over its CAPM-implied equilibrium return, with view confidence set from how sharp the posterior is) and apply the standard closed-form Black-Litterman update (`oilbn.models.black_litterman.BlackLittermanAllocator`) to a simple two-asset WTI/S&P 500 universe. The near-certain bullish view shifts WTI's posterior expected annual return from the 20.6% equilibrium estimate to 32.5%, and the resulting unconstrained mean-variance-optimal weight on WTI rises from an equal-weighted 50% to 81.8%. This is intentionally a mechanism demonstration, not a backtested strategy: a genuine allocation decision would need the full multi-asset opportunity set, transaction costs, and the out-of-sample forecast-accuracy caveat from Section 1.2 built into the view-confidence calibration (an overconfident view derived from a model that only achieves 70% directional accuracy would be a modeling error).

2 Critical Assessment of Alvi's Claimed Contributions

Alvi's dissertation (Chapter 5, Conclusions & Future Work) (Alvi) closes with eight claimed contributions. Each is assessed below against what this project was actually able to replicate and observe first-hand, not merely against the original text's own framing.

1. **Replacing EGARCH-M-derived views with Bayesian-model- derived views for Black-Litterman.** *Achieved, and now demonstrated end-to-end in Section 1 of this report.* Alvi's own dissertation proposes this substitution as future work rather than implementing it; our `BlackLittermanAllocator` demonstration is, to our knowledge, a working instantiation of exactly this idea. It matters because it replaces a purely statistical volatility-forecasting input (EGARCH-M) with a structurally-interpretable, auditable one (a Bayesian network's posterior), directly addressing the "black box" objection a risk committee would reasonably raise against a pure time-series model.

2. **Using HMM-discretized time series as inputs to belief networks is novel.** *Partially achieved and independently confirmed useful, but "novel" is an overclaim.* We independently re-derived the same two-stage design (regime-detect, then learn structure over the discretized regimes) without reference to Alvi's implementation details, and found it necessary for the same reason Alvi likely did: a belief network cannot consume continuous, non-stationary price levels directly. However, combining HMM state-decoding with downstream structure learning predates this 2018 dissertation in the broader regime-switching literature; the genuine contribution is the specific *application* to crude oil risk management, not the general technique.

3. **A working, directly-deployable trading mechanism capable of independent decision-making.** *Not achieved, by our own replication.* Section 1's honest finding – 69.7% one-month-ahead regime accuracy, below the 87.9% naive-persistence baseline – directly contradicts "deployable"; a mechanism that underperforms the trivial baseline cannot be handed an independent capital-allocation mandate without a human overseeing it. This matters because it is exactly the kind of overclaim a risk manager must be trained to catch before signing off on a model.

4. **Structure-learning-based, so near-autonomous and requiring no prior expert knowledge beyond dataset selection.** *Achieved, with one caveat.* Our Hill-Climbing pipeline requires no manually-specified edges, confirming this claim; the one qualification is that *dataset selection itself* is not a trivial, expert-knowledge-free step – deciding to discretize financial series via regime-detection but macro/geopolitical series via simple terciles (Section 4 of the GWP2 report) is itself a domain judgement call that materially shapes what the algorithm can possibly discover.

5. *(Duplicate of contribution 2 in the source text, verbatim – not independently assessed.)*

6. **Python PGM libraries' abstraction level aided theoretical understanding of the design process.** *Achieved, and we concur directly.* Implementing the four HMM algorithms from scratch (GWP2) while using `pgmpy`'s higher-level Hill-Climbing/BIC/CPD abstractions for the belief-network side gave us the same pedagogical contrast Alvi describes: writing Baum-Welch by hand forces an understanding of *why* EM must be used (Section "Hidden Markov Models: A Latent-Variable-Model Definition" of the GWP2 report), while `pgmpy`'s `HillClimbSearch` lets attention shift to interpreting the *result* (edge economic plausibility, Markov blanket contents) rather than the search mechanics.

7. **Provides forecasts via a systematic, event-driven global-macro strategy with higher-return potential than hedge-fund/fixed-income strategies.** *Not supported by our replication.* This is the single most over-reaching claim in the list: no backtested, cost-adjusted return series is presented anywhere in the original dissertation to substantiate a hedge-fund-beating return claim, and our own forecast-accuracy result (Section 1) actively argues against treating the model's regime calls as a standalone systematic strategy without further validation.

8. **Helps energy policy makers understand oil-market structure to draft better policy.** *Plausibly achieved, and the strongest-surviving claim in our view.* Regardless of the model's forecast accuracy, the *structure* it recovers – which variables belong in `wti_regime`'s Markov blanket, and which do not – is itself informative to a policy audience: our replication across three different samples consistently finds that an industrial-production indicator and the Brent benchmark, not pure financial-market variables, are the proximate drivers of the WTI regime. A policymaker monitoring physical-market indicators rather than financial-market sentiment to anticipate oil-price regime shifts is a defensible, model-structure-derived recommendation that does not depend on the forecast-accuracy shortfall documented above.

In summary, this project's replication supports Alvi's *structural* claims (contributions 1, 2, 4, 6, 8) considerably more than his *performance* claims (contributions 3 and 7): the belief-network machinery reliably recovers an interpretable, economically plausible, and reasonably stable causal structure without expert input, but the resulting model's one-month-ahead forecast accuracy does not, on this replication, clear the bar of a naive persistence baseline – a result future work (and genuine commercial deployment) would need to address directly rather than assume away.

3 Discussion: What This Means for an Investment Decision

This section is written for a reader who needs to decide whether to act on this project's output, not for a reader who needs to audit its mathematics.

3.1 What the model actually tells you

The headline finding of this project is simple: a computer program, given only historical prices and a handful of publicly available industry statistics – and given *no* instruction about which of those statistics matter – consistently figures out on its own that crude-oil price swings are tied most closely to *physical, industrial* activity (how much the oil-and-gas-support sector of the economy is growing or shrinking) and to the *other* major benchmark crude price (Brent), and only loosely, if at all, to broad financial-market sentiment (the stock market, the dollar, bond yields). That is a genuinely useful piece of information for anyone trying to understand *why* oil prices move, independent of whether the model can also tell you *which direction* they will move next.

3.2 What the model does not (yet) tell you

Separately, and just as important: when we asked the same model to make a concrete, one-month-ahead call on whether oil would be in a rising, falling, or flat period, it was right about 7 times out of 10 – but simply assuming "next month looks like this month" was right closer to 9 times out of 10. In plain terms: on this test, the model's market-timing signal was not yet good enough to act on by itself. This is a common and important finding in quantitative investing more generally – many signals that look statistically interesting completely fail to beat the simplest possible comparison (just assuming things stay the same), and the discipline of checking against that simple comparison, rather than skipping straight to "the model found something", is exactly what separates a credible research process from an overconfident one.

3.3 What we would recommend to a decision-maker

- **Use the model's structural finding as a monitoring tool, not a trading signal.** Knowing that industrial activity and the Brent benchmark are the variables to watch is actionable today, at no risk: it tells a risk manager or a policy analyst *where* to focus limited attention, even without trusting the model's specific price-direction calls.
- **Do not allocate real capital based on the current forecast alone.** A trading or hedging decision sized directly off this model's monthly regime call would, on the evidence gathered here, likely underperform the much simpler rule of just assuming "no change." Before any

capital is put behind this specific forecast, it would need to clear the same bar in a longer, independent test period, and ideally be combined with other signals rather than used alone.

- **The portfolio-tilting demonstration (Section 1) illustrates a mechanism, not a recommendation.** We showed that a confident model view *can* be translated into a concrete portfolio tilt (shifting an equal-weighted position toward oil when the model is bullish). That machinery is sound and reusable, but plugging in today's specific, not-yet-validated forecast would be putting a mechanism ahead of the evidence that mechanism should be fed by.
- **Cost-effectiveness.** Every input used by this project (Yahoo Finance prices, FRED and EIA public statistics) is free and automatically refreshable, and the entire pipeline runs in minutes on an ordinary laptop. The honest 70%-vs-88% comparison above cost nothing extra to produce and should be considered a required, essentially free quality check before any such model is trusted with a real decision – there is no excuse, cost-wise, for skipping it.

In short: this project demonstrates a transparent, inexpensive, and genuinely interpretable way to monitor what is driving the oil market, and a clean mechanism for turning a model view into a portfolio decision – but it also demonstrates, honestly, that the specific market-timing signal produced by this particular model, on this particular sample, is not yet strong enough to deploy on its own.

Works Cited

Alvi, Danish A. "Application of Probabilistic Graphical Models in Forecasting Crude Oil Price". 2018. U College London, Department of Computer Science, Bachelor's thesis. arxiv.org/abs/1804.10869. Accessed 1 Jan. 2024.